

Challenge 1 Circle Spiral Input

CircleSpiralInput.py - Challenge Problem

import turtle # set up turtle graphics

t=turtle.Pen()

t.speed(0)

turtle.bgcolor('black')

Set up a list of any 8 valid Python color names

colors=['red', 'yellow', 'blue', 'green',

'orange', 'purple', 'white', 'gray']

Ask the user for the number of sides, between 1 and 8, default of 4

sides = int(turtle.numinput("Number of circles",

"How many circles do you want (1-8)?", 4, 1, 8))

Draw a colorful spiral with the user-specified number of circles

for x in range(180):

t.pencolor(colors[x % sides]) # only use the right number of colors

t.circle(x)

t.left(360 / sides + 1) # turn 360 degrees / # of sides, plus 1

t.width(x * sides / 200) # make the pen larger as it goes outward